

MISP

Malware Information Sharing Platform & Threat Sharing

Research Report

Advanced Monitoring System — Group Project

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1 Introduction

This report aims to present an analysis of MISP (Malware Information Sharing Platform and Threat Sharing), the primary goal of this research is understanding what MISP is, how it works, what features it offers. The information provided is based on the official MISP website, some GitHub documentation, and a few academic articles.

2 What is MISP?

2.1 Overview

MISP is an open-source threat intelligence platform used for collecting, storing, and sharing cybersecurity threat data, such as Indicators of Compromise (IoCs). The problem MISP addresses is the inefficient sharing of threat data through emails and PDFs, which makes automated processing difficult.

MISP allows threat data to be stored in an organized way, such as malicious IP addresses, file hashes, and domain names. It also supports sharing more complex data, such as fraud indicators and vulnerability reports. Over the time, MISP has become a more comprehensive tool for various cybersecurity needs.

2.2 Who Uses It?

MISP is used by different professionals in cybersecurity. This includes Security Operations Center (SOC) analysts, incident responders, malware researchers, fraud analysts, and law enforcement agencies. Key organizations such as NATA, national CERTs (Computer Emergency Response Teams) across Europe, financial institutions, ISACs (Information Sharing and Analysis Centers), and security vendors.

3 How MISP Works

3.1 Architecture

MISP is a web-based application built using PHP with the CakePHP framework, and it uses MySQL/MariaDB for its database. It can be deployed in various environments: on-premises, in the cloud, or as a hosted service. One key feature of MISP's architecture is that individual MISP instances can run independently but still synchronize with each other through push/pull mechanisms.

The main components of MISP include:

- A web interface for analysts to manage and interact with threat data.
- A REST API for programmatic access (documented in OpenAPI format).
- A correlation engine that automatically links related indicators of compromise (IoCs).
- A synchronization system that facilitates sharing between multiple MISP instances.
- A Python-based module system for extending functionality.
- Real-time event notifications using ZMQ/Kafka channels.

3.2 Data Flow

Data in MISP flows through a structured process:

1. **Ingestion** — Data enters MISP through various methods, including manual entry, bulk imports (such as CSV format), automated feeds, synchronization from other MISP instances, or API calls.
2. **Structuring** — Once ingested, the data is organized into events. Events are tagged with relevant taxonomies, and the correlation engine links new data with existing indicators.
3. **Analysis** — Analysts review the correlations, add context, and collaborate using features like proposals and comments.
4. **Sharing** — Events are shared according to predefined rules, which could be organization specific, community wide, or custom sharing groups.
5. **Consumption** — External tools (such as IDS, SIEMs, and SOAR platforms) consume the data by either exporting it or accessing it through the API.

3.3 Deployment Options

MISP can be deployed in several configurations depending on the organization's needs:

- **Single Instance** — A single server used internally within an organization.
- **Hub and Spoke** — A central MISP instance syncs data with multiple satellite instances.
- **Federated Mesh** — Multiple MISP instances sync data with each other in a peer-to-peer model.
- **Hybrid** — An internal instance is used for sensitive data, while an external one is used for sharing data with the broader community.

4 Core Features

MISP offers several key features that make it a powerful tool for threat intelligence sharing. These features include event management, data correlation, advanced sharing options and automation tools.

4.1 Events, Attributes, and Objects

The fundamental unit in MISP is an event, which represents a specific cybersecurity incident or threat. Each event contains attributes, such as IP addresses, file hashes, or domain names, which represent specific indicators of compromise (IoCs). MISP supports over 170 types of attributes, grouped into 16 categories.

For more complex data, MISP uses objects to group related attributes together. For instance, a "file" object might include the filename, various hash values, and file size as a single unit. There are over 300 object templates available to organize data.

4.2 Taxonomies and Galaxies

MISP supports taxonomies, which are standardized tagging systems for categorizing threat data. These taxonomies help maintain consistency in data labeling across different organizations. Some examples include:

- TLP (Traffic Light Protocol)
- MITRE ATT&CK
- ENISA classifications
- NATO markings

- GDPR categories

Additionally, galaxies provide curated knowledge bases of related entities, such as threat actors, malware families, or attack techniques. These can be linked to events to provide richer context.

4.3 Correlation Engine

The correlation engine is one of MISP's most powerful features. When a new attribute is added to the system, the engine automatically checks it against existing data and identifies related indicators. The engine supports various matching techniques, including exact matches, fuzzy hashing (using ssdeep), and CIDR block matching for IP ranges.

4.4 Sharing and Synchronization

A primary purpose of MISP is to facilitate sharing and synchronization of threat intelligence. Data can be shared at various levels of granularity:

- Within a single organization
- With a community of trusted partners
- Through custom sharing groups

MISP allows sharing control at the individual attribute level, ensuring sensitive information can be shared securely while protecting critical data.

4.5 API and Automation

MISP provides a robust REST API, allowing programmatic access to its features. It also includes PyMISP, a python library for easy interaction with the API. Additionally, MISP supports workflow automation, enabling tasks such as data validation, enrichment, and publication approval to be automated.

4.6 Feeds

MISP can ingest external threats feeds, like the CIRCL (Computer Incident Response Center Luxembourg) OSINT feed, and can also generate feeds for other systems to consume. These feeds can be scheduled to pull data at specific intervals, ensuring up-to-date information is always available.

4.7 Other Notable Features

- **Sightings:** This feature allows organizations to confirm that they have observed a shared indicator in their environment.
- **RBAC (Role-Based Access Control):** Configurable roles and permissions allow organizations to control who can access or modify specific data.
- **Dashboards:** MISP includes drag-and-drop functionality for building visual dashboards that display threat intelligence in an easily understandable format.
- **Warning Lists:** MISP includes filters for known false positives, such as common IPs from services like Google DNS and Cloudflare.

5 Use Cases

MISP is a versatile platform used in various sectors to enhance threat intelligence sharing and improve response to cybersecurity incidents. Below are some of the key use cases where MISP plays a critical role.

5.1 SOC and Incident Response

In Security Operations Center (SOC), MISP serves as a backend for managing threat intelligence. Indicators of Compromise (IoCs) stored in MISP are automatically converted into IDS (Intrusion Detection System) rules for tools like Snort or Suricata and pushed to SIEMs (Security Information and Event Management systems). This allows SOC teams to quickly identify and respond to threats.

Incident Response Teams (CSIRTs and CERTs) also use MISP during security incidents. It helps them document findings, correlate with historical data, and share threat intelligence with peer organizations via synchronized communities.

5.2 Financial Sector

MISP has a dedicated community for the financial sector managed by CIRCL (Computer Incident Response Center Luxembourg). In this community, banks and payments processors share threat data related to phishing, credential theft, and payment fraud. MISP provides specialized attribute types for financial data, such as IBANs, and credit card numbers, helping organizations protect sensitive financial information.

5.3 Military and Government

NATO uses its own instance of MISP to share defense-related threat intelligence among member nations. National CERTs also utilize MISP to strengthen their cybersecurity efforts and collaborate with other governments on various threats. This collaborative environment helps ensure that military and government entities can efficiently exchange relevant data.

6 Technical Details

MISP's technical infrastructure is designed to handle complex threat intelligence data. It uses structured formats and integrates seamlessly with various security tools.

6.1 Data Model

The MISP data model is formalized as an IETF Internet-Draft, which specifies how data is represented and exchanged within the platform. The key components of the data model include:

- **Events:** Represent incident or threats reports.
- **Attributes:** Indicators of compromise (IoCs) such as IP addresses, domain names, or file hashes.
- **Objects:** Groupings of related attributes, such as files or network traffic.
- **Tags:** Metadata used for categorizing and organizing data.
- **Galaxy Clusters:** Curated sets of related entities such as threat actors, malware families, and attack techniques.
- **Sightings:** Confirmations that an organization has observed a specific IoC.

All data is represented in JSON format, and it is accessible via the MISP API, making it easy

to interact with.

6.2 Supported Formats

Table 1 summarizes the formats MISP can import and export.

MISP JSON	Native format for synchronization and API
STIX 1.x / 2.x	Standard threat intelligence exchange format
Snort / Suricata	Auto-generated IDS rules
Bro / Zeek	Network monitoring rules
OpenIOC	Mandiant/FireEye ecosystem compatibility
CSV / text	Simple exports for any tool
YARA / Sigma	Malware detection and SIEM rules
RPZ	DNS-based blocking

Table 1: MISP supported formats

6.3 Integrations

MISP integrates with a variety of security tools to extend its functionality and enable automated workflows. Some notable integrations include:

- **SIEMs:** Such as Splunk (via MISP42 app), Elastic, Microsoft Sentinel, and QRadar.
- **SOAR platforms:** Tools like TheHive/Cortex and XSOAR for security automation and orchestration.
- **Case Management:** MISP integrates with tools like FlowIntel, TheHive, and Request Tracker to facilitate incident tracking and case management.
- **Enrichment:** MISP integrates with enrichment services such as VirusTotal, Shodan, PassiveTotal, and CIRCL Passive DNS.
- **Visualization:** MISP supports integration with visualization tools like Maltego and Jupyter notebooks for data analysis and reporting.

7 Community and Ecosystem

7.1 Sharing Communities

One of the things that sets MISP apart from other tools is the size of its community. Table 2 lists some of the known public communities.

CIRCL MISPPRIV	Private sector	1,100+ organizations
CIRCL Financial	Finance	Banks and payment processors
NATO MISP	Defence	NATO member nations
FIRST MISP	Multi-sector	Global incident response community
MISP-LEA	Law enforcement	International LEA sharing
ICS-CSIRT.io	Industrial	ICS/OT threat data

Table 2: Selected public MISP sharing communities

There are also many private communities that are not publicly listed.

7.2 Standards and Governance

MISP follows an open-source, community-driven governance model, ensuring that no single organization can control or commercialize the platform. The MISP data formats are formalized as IETF Internet-Drafts, which ensures that the data exchanged within MISP can be used by other tools as well. This guarantees that MISP is not tied to a specific software and enhances its flexibility within the broader cybersecurity ecosystem.

The project is led by CIRCL (Computer Incident Response Center Luxembourg), with funding support from the EU, and the development follows open-source principles that encourage community participation.

8 Security and Privacy

Sharing threat intelligence inherently involves concerns about the sensitivity and privacy of the data being exchanged. MISP takes various steps to mitigate these concerns and ensure that shared information remains secure. Below are some of the key security and privacy features of MISP:

- **TLP Tagging (Traffic Light Protocol):** MISP uses TLP to classify the sensitivity of the shared data. The four levels of TLP are:
 - **RED:** Restricted to a very limited audience.
 - **AMBER:** Shared with caution, typically for specific communities.
 - **GREEN:** Widely shared but not public.
 - **CLEAR:** Public data.
- **PAP Tagging (Protection Against Publication):** This mechanism defines what recipients can do with the shared data, ensuring that it is not published or redistributed beyond the agreed-upon audience.
- **Granular Distribution Controls:** MISP allows distribution control down to the individual **attribute level**. This enables users to share specific pieces of data selectively, providing fine-grained control over sensitive information.
- **Sharing Groups:** Sharing groups allow organizations to define specific audiences for their threat data, ensuring that sensitive information is only shared with trusted partners or groups.
- **Delegated Sharing:** This feature enables **pseudo-anonymous publication** of data. It

allows organizations to share data without revealing their identity, thus protecting the privacy of the contributor.

- **Encryption:** MISP supports encryption for sensitive notifications. It uses **GnuPG** and **S/MIME** encryption methods for email notifications, adding an additional layer of protection for shared intelligence.
- **MISP-Guard:** This is a security feature designed to prevent the leakage of sensitive information. It acts as a safety net for organizations to avoid inadvertently sharing overly sensitive data.

MISP also provides compliance guidance for various regulatory standards, including GDPR, ISO/IEC 27010:2015, and the EU NIS Directive, ensuring that organizations remain compliant while sharing threat intelligence.

9 Strengths and Limitations

9.1 Strengths

From the research, the primary advantages of MISP are:

- **Free and Open Source:** MISP is completely free to use and open-source, offering a strong license guarantee that ensures long-term accessibility and flexibility.
- **Best-in-Class Sharing and Synchronization:** MISP excels in sharing and synchronizing threat intelligence data across multiple organizations and communities. The platform supports granular control over the distribution of sensitive data.
- **Extensive Ecosystem:** MISP has a large ecosystem with over 150 taxonomies, 300+ object templates, and many modules, making it a highly adaptable platform for a wide range of cybersecurity needs.
- **Active Community:** MISP has a large and active community of users and contributors, providing valuable feedback and continuous improvement of the platform.
- **Standard Format Support:** MISP supports widely recognized formats like STIX, in addition to its own well-documented format, ensuring interoperability with other threat intelligence platforms.
- **Flexible Deployment Options:** MISP can be deployed in a variety of environments, including on-premises, in the cloud, or as a federated instance, allowing organizations to choose the best option for their needs.

9.2 Limitations

While MISP offers numerous advantages, it also has certain limitations that organizations should consider:

- **Setup and Maintenance Effort:** Setting up and maintaining an instance of MISP requires significant effort, particularly for larger deployments, as it involves system administration skills and knowledge.
- **Performance at Scale:** MISP's performance can degrade when dealing with large volumes of data unless proper database tuning and optimization are applied.
- **Steep Learning Curve for New Users:** The user interface (UI) of MISP can be challenging for new users, and it requires time and experience to fully understand and use the platform effectively.

- **Data Quality Dependent on Community Contributions:** The quality of the data in MISP depends entirely on what the community contributes. Poor data quality can lead to ineffective threat intelligence sharing.
- **Limited Advanced Analytics:** While MISP is great for managing and sharing threat intelligence, it lacks advanced analytics features such as machine learning and behavioral analysis, which require integration with external tools.
- **Outdated Technology Stack:** MISP is built using PHP and the CakePHP framework, which some consider outdated compared to modern web development technologies.

9.3 Comparison

Table 3 shows how MISP compares to some alternatives.

Cost	Free	Free	Free	Paid
Sharing	Excellent	Good	Limited	Varies
STIX	1.x + 2.x	Native 2.1	Basic	Usually good
Community	Very large	Growing	Large	Vendor-tied
Maturity	Since 2011	Since 2019	Since 2016	Varies

Table 3: MISP vs. alternative platforms

10 Conclusion

MISP is a mature and widely-adopted platform that excels in structured threat intelligence sharing. It is particularly effective as a threat intelligence layer in an advanced monitoring system, where it can feed Indicators of Compromise (IoCs) into Intrusion Detection Systems (IDS), Security Information and Event Management (SIEM) platforms, and other security tools. Additionally, MISP facilitates collaboration with external partners, which enhances the collective defense against emerging threats.

One of the main strengths of MISP is its flexibility, providing a rich ecosystem with a wide variety of taxonomies, object templates, and integration options. The combination of these features, along with an active community, makes MISP one of the best open-source options for threat intelligence sharing and collaboration.

However, MISP's operational complexity should not be underestimated. Setting up and maintaining the platform requires dedicated effort, especially at scale. The user interface (UI) can be difficult for new users to navigate, and performance may degrade with large datasets unless careful attention is paid to database optimization.

Despite these challenges, MISP remains the most practical choice for organizations looking to implement an open-source threat intelligence sharing platform. Its strong ecosystem, support for standard formats, and active community make it a valuable tool for cybersecurity teams across various sectors.

Overall, MISP offers a comprehensive and flexible solution for sharing, managing, and analyzing threat intelligence, and it is highly recommended for organizations that require secure, scalable, and collaborative threat intelligence sharing capabilities.

11 References

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